

Muhammad Abdullah Khan

Software Engineer | Applied AI & Data Science Engineer

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SUMMARY

Applied AI Engineer focused on **fine-tuning open-source machine learning models** for high performance on **constrained hardware**. Highly experienced in **end-to-end data science workflows**, **algorithmic data preprocessing**, and building **autonomous computer vision agents**. Combines a deep understanding of **complex data pipelines** and **core ML optimization** with the backend engineering skills needed to deploy **fast, reliable AI infrastructure**.

PROFESSIONAL EXPERIENCE

QUBIC Web3 QA Analyst	11/2023 – 03/2024 Remote
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EDUCATION

COMSATS University Islamabad Bachelor of Science in Software Engineering (BSSE)	09/2023 – Present Lahore, Pakistan
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SKILLS

Agentic Development & Orchestration Multi-Agent Architectures, Agentic Prototyping (Claude Code, Codex, Antigravity) .LangGraph, MCP Integration, AI-Driven CI/CD, Autonomous Workflow Automation LLM Orchestration & Safety Engineering Local Model Deployment (Ollama, llama.cpp), Multi-Model API Routing, Enterprise AI Guardrails, Prompt Injection Defense, PII Redaction, Hallucination Mitigation Strategies.	ML Infrastructure & Data Engineering Python, FastAPI, RESTful Model Serving, Database Management (PostgreSQL), Secure Proxy Architecture, High-Availability Backend Scaling, ML Pipeline System Architecture. Core Machine Learning & Computer Vision PyTorch, Parameter-Efficient Fine-Tuning (PEFT/LoRA), Model Quantization, Vision-Language-Action (VLA) Models, Visual Grounding, Algorithmic Data Preprocessing pipelines.	Applied NLP & Semantic RAG Retrieval-Augmented Generation (RAG), Semantic Search & Indexing, Text Embedding Pipelines, Vector Databases (ChromaDB, LanceDB), Context Window Management, Abstract Syntax Tree (AST) Parsing. Software Engineering System Architecture, Requirements Analysis, Legacy Re-engineering, Code Optimization & Refactoring, Quality Assurance, Full-Cycle SDLC.
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PROJECTS

OmniVLA

01/2026 – Present

Hardware-Constrained Autonomous GUI Agent

- Built a localized LLM agent optimized for low-VRAM environments, utilizing UI tree parsing and coordinate grounding for automated screen navigation and task automation.
- Developed a dedicated, responsive web interface to effectively showcase the model's capabilities, hardware performance metrics, and technical documentation.

VigilAI Proxy

04/2026 – 05/2026

LLM Security Gateway & Telemetry

- Architected a local LLM security gateway using Python and FastAPI to sanitize AI prompts, filter model outputs, and route multi-provider API requests (OpenAI, Claude, Gemini).
- Built a responsive, data-driven frontend dashboard using Vanilla HTML/CSS/JS to visualize real-time API telemetry, token costs, and backend execution latency via interactive Chart.js modules.

OpaqueCI

05/2026 – 06/2026

AI Powered Zero-Trust Code Review Agent

- Engineered an on-premise AI agent that automates security and style audits for Git pull requests utilizing semantic RAG pipelines (LanceDB) and local LLMs.
- Integrated Tree-sitter for Abstract Syntax Tree (AST) parsing to feed complex structural code context into the AI model.
- Created a monochromatic, minimalist web console using custom CSS to monitor live execution logs, manage rule libraries, and render vulnerability distributions.

Stride

01/2025 – 12/2025

AI-Driven Physiological Tracking System

- Built a highly responsive, animated frontend application leveraging Flutter and Riverpod to deliver a seamless, high-performance cross-platform user experience.
- Integrated generative AI capabilities by combining custom machine learning models with the Gemini API to autonomously generate and serve dynamic routines.
- Deployed a robust serverless architecture using Firebase and Dockerized ML microservices, ensuring fast, scalable execution of AI model inferences.

ShiftsSaga

01/2026 – 02/2026

Full-Stack CRM & Landing Page

- Designed and deployed a highly responsive, modern landing page optimized for lead generation and user engagement.
- Architected a complete backend CRM admin panel to manage user data, track interactions, and streamline internal operations.
- Utilized agentic coding workflows (Antigravity, Codex) to rapidly prototype frontend components and accelerate backend logic integration.